

Comparing AI-Generated and Human-Crafted T-Shirt Layouts through an XAI Lens

Key Design Elements and Implications for Co-Creative Tools

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Abstract

In recent years, the apparel industry has also begun to draw attention to the use of generative AI for design assistance. This study utilized 30 existing T-shirt illustrations provided by graniph and compared: (1) the layouts proposed by the generative AI system “Gemini,” (2) the actual product layouts, and (3) the choices made by five professional designers. First, when evaluated based on eight strictly defined categories (sub-categories), Gemini’s layouts matched the actual product in only 12 of the 30 cases. However, by grouping similar placement categories into five broader categories, the number of matches increased to 20. Furthermore, in terms of these broad categories, the matches between Gemini and the five designers ranged from 18 to 24, indicating that Gemini’s proposals exhibit a certain level of commonality with human decisions. In addition, when Gemini explained in text why it chose a particular layout, over half of the illustrators found these explanations “reasonable,” suggesting that explainable AI can effectively support collaborative design. This study clarifies both the practical and academic significance of using generative AI in the fashion design field and highlights the importance of interaction design and explanation methods to facilitate smoother collaboration between humans and AI.

CCS Concepts

• Human-centered computing; • Interaction design; • Empirical studies in interaction design;

Keywords

Generative AI, Human-AI Collaboration, Interaction Design, Layout Recommendation

ACM Reference Format:

Miki Katsuragi and Kenji Tanaka Prof.. 2025. Comparing AI-Generated and Human-Crafted T-Shirt Layouts through an XAI Lens: Key Design Elements and Implications for Co-Creative Tools. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3706599.3720016>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3720016>

'25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3706599.3720016>

1 Introduction

In the fashion industry, there is a need to propose designs quickly in response to shifting consumer needs and trends. In recent years, advances in image generation AI and natural language processing AI have drawn attention to the potential of AI-based support tools in design. In particular, T-shirts—offering a relatively simple canvas—have attracted notice as an area where it is easy to automatically generate diverse style and layout proposals. However, there remains a challenge: given real-world constraints such as brand concepts, manufacturing processes, and consumer preferences, the layouts proposed by AI do not necessarily align with practical operational requirements.

Beyond fashion, the fusion of AI-based automatic proposals with the sensibilities and expertise of human specialists is an important topic in many creative fields. While AI can rapidly present a wide range of patterns, brand identity, decisions rooted in sales experience, and subtle nuances of trends rely heavily on the knowledge of professional designers. Moreover, because the internal workings of generative AI can easily become a “black box,” designers may be reluctant to adopt AI proposals if they cannot understand the reasons behind them. Consequently, there is growing anticipation for Explainable AI (XAI), which aims to clarify the perspectives from which AI selects a given layout.

In this study, we used 30 existing T-shirt illustrations from the apparel brand graniph and had the generative AI model “Gemini” propose layouts for them. We then made a quantitative comparison with: (1) the actual configurations used in production, and (2) layout selections made by five professional designers, in order to evaluate how well Gemini’s proposals align with human judgment and real-world considerations. Furthermore, we presented illustrators with the textual explanations of “why Gemini chose each layout” to explore whether Explainable AI can foster collaboration in the fashion design realm.

In our experiments, when evaluating according to eight strict categories (such as a small front placement, a point placement, a diagonal lower placement, etc.), the agreement rate was by no means high—only 12 out of the 30 items matched. However, when we broadly grouped similar layouts (e.g., treating front small-to-medium size as “front,” and large front or back placements as “large”), the number of matches with actual products rose to 20, suggesting that there is some degree of correlation between layouts chosen by Gemini and those chosen by human designers. Moreover,

multiple illustrators highly rated the text explanations of “why a particular layout was chosen” offered by Gemini as “reasonable,” indicating that providing rationales may increase designers’ acceptance and understanding.

The contributions of this study can be summarized as follows.

- Comparison of AI-generated layouts and human experts: In the concrete application field of T-shirt design, this study provides a case analysis that comprehensively compares generative AI, professional designers, and existing products.
- Verification of Explainable AI’s effectiveness: By measuring expert reactions and evaluations after being presented with explanations such as “Why was this particular design chosen?”, the study demonstrates that explainability is a critical factor in gaining acceptance of AI-generated proposals.
- Practical implications for the fashion sector: Since broadly grouping the layout categories raises the match rate and the process by which designers integrate brand context remains indispensable, the study offers guidelines for future implementation of generative AI in design.

2 Related Work

The use of generative AI in the fashion domain has recently been explored in numerous ways, including integration with e-commerce. For instance, [1] examines Midjourney as a case study and illustrates how the diversity and speed of AI-based visual generation can expand conventional fashion-design workflows. At the same time, the fashion industry involves a wide range of considerations—such as maintaining brand image, responding to changing consumer tastes, and dealing with physical constraints of printing and sewing—which the AI must also take into account. Research using Generative Adversarial Networks (GANs) to generate fashion designs has proposed various approaches, including the automatic creation of clothing patterns and prints [2] and adversarial-network-based fashion design [3].

While these studies efficiently generate a variety of design ideas tailored to consumer preferences, they are frequently criticized for not sufficiently reflecting industry-specific constraints (e.g., sewing techniques, fabric cost, and size variations). Moreover, the challenge of modeling the unique brand worlds and subtle trend nuances cultivated by fashion labels has been identified as another major issue [4]. In [5], the authors investigate how artificial intelligence (AI) impacts the fashion-design process. They discuss ways in which generative algorithms can produce new design concepts and encourage human designers to explore fresh possibilities.

Looking specifically at the topic of “layout,” prior research has examined AI-driven layout optimization for fields such as advertising banners and magazine design [6, 7]. These approaches frequently involve machine learning or heuristic algorithms to automatically generate designs, taking into account image balance and the guiding of the viewer’s eye [8]. Although some studies propose combining text content with images to suggest layouts [9], there are relatively few studies that have systematically examined how closely these AI-proposed designs align with the judgment of human experts.

Explainable AI (XAI) has attracted considerable attention across a wide range of fields in recent years. In particular, for tasks that

support expert decision-making, researchers have noted the importance of presenting explanations about “why the AI made a given proposal or prediction” in a human-understandable way [10, 11]. For instance, in medical diagnosis and financial risk assessment, introducing XAI has been reported to alleviate experts’ anxiety or reluctance toward using AI outputs, thereby enhancing trust [12]. Meanwhile, although AI support is increasing in creative fields such as fashion design and art, if it remains unclear why the AI considers a particular design to be superior, experts may find it difficult to judge how much to trust an AI-generated idea [13].

Beyond the fashion industry, other research has compared AI-based outputs to those of human designers in various design tasks. For example, Joosten et al. [14] discuss how generative AI can outperform human ideation in originality under certain conditions, whereas [15] report that AI-suggested ideas might increase design fixation and reduce overall ideation diversity. In architecture and engineering, [16] highlight the need for upskilling professionals who seek to integrate AI-driven workflows. [17] focuses on graphic design education, emphasizing the importance of understanding ethical and creative implications when students adopt generative AI tools. Additionally, [18] explore the aesthetic judgment factors that distinguish AI- from human-generated art, suggesting a persistent bias favoring perceived human-made content. In the context of fashion design itself, [19, 20] find that designers can leverage AI outputs effectively for early-stage concept ideation, although brand alignment and intellectual property concerns remain. Finally, from a broader design management perspective, [21] argues that organizations capable of strategically deploying generative AI can gain a competitive edge in product development and user experience.

As noted above, most existing research addresses one of three main topics: (1) the foundational potential and challenges of AI in design, (2) general methods of layout optimization, or (3) the social and philosophical debates surrounding Explainable AI. In contrast, our study connects these perspectives in the concrete application area of T-shirt design, distinguishing itself by comparing and evaluating three entities: actual product layouts, the layout choices of five professional designers, and the proposals from the AI model (Gemini). Moreover, by incorporating an XAI perspective through assessments of the AI’s textual explanations, we provide empirical insights into methods for co-creative work with AI.

- RQ1: How closely do the layouts chosen by Gemini match the actual product layouts (“production”) and those selected by professional designers?
- RQ2: When we compare the layout in smaller, more detailed categories (8 types) versus broader categories (5 types), how do the match rates and AI-human agreement levels change?
- RQ3: How do Gemini’s textual explanations for “why this layout was chosen” affect the sense of acceptance and evaluation among designers and illustrators, and can these explanations promote creative collaboration?

3 METHODOLOGY

This section provides an overview of the data used in this study (T-shirt illustrations), the definition of layout categories, the settings for the generative AI model (Gemini), and the procedure for the designer evaluation experiment.



Figure 1: Examples of each T-shirts layout

3.1 T-Shirt Illustrations and Layout Categories

In this study, we used 30 T-shirt illustrations actually commercialized by the graniph company. These illustrations cover a wide range of designs, including animal motifs, characters, and text elements, and they vary in color and size. Since layout selection is strongly influenced by the shape and feel of each illustration, we took care to avoid bias toward any single style. In consultation with designers at graniph, we defined eight categories for print positions on a T-shirt—[front-small, front-middle, front-large, point (chest icon), angled-lower, neck-area, back, other]—as well as five broader categories—[front, large front/back, angled-lower, neck-area, other]. In Figure 1, the black text corresponds to the eight categories, while the blue outlines represent the five broader categories.

graniph uses its own unique naming for layouts internally, but to minimize discrepancies between AI suggestions and human interpretations, we replaced these with more general terms. For instance, “front-small” refers to a relatively small placement in the center area, and “point” indicates an icon-like placement on the chest.

3.2 Generative AI (Gemini) Setup

Gemini is a Google’s multi modal generative AI which can recognize images with conversational capabilities. It can propose optimal layouts and supplemental explanations based on a given illustration and text prompt. For this experiment, we used the Gemini API (gemini-1.5-pro-002) and, for each illustration, provided Gemini with the instruction: “Which part of the T-shirt is the best placement for this illustration? Please choose one of the eight categories and explain the reason.”

Example instruction:

“Given the following T-shirt illustration, choose from {front-small, front-medium, front-large, point, angled-lower, neck-area, back, other}. Which layout is best, and why?”

Output format:

- **Layout category:** One category

- **Explanation (Why?):** A few lines of text explaining why Gemini chose this category. We used these generated reasons later when evaluating illustrators’ reactions.

We ran this process for all 30 illustrations, providing each one to Gemini in turn and recording the outputs.

3.3 Human Designers’ Evaluation

We recruited five designers (p1 to p5, aged 25–40, mixed gender) from graniph, each with at least three years of experience in T-shirt design. They are actively involved in T-shirt design, understand the brand’s concepts, and are knowledgeable about printing techniques. We also asked these same designers to assess the validity of Gemini’s explanation texts.

For the experiment, each of the five designers (p1 to p5) took part in a survey where they viewed the same 30 illustrations and answered the following question, choosing one of the eight layout categories they felt was most appropriate as below:

“For each illustration, please choose the layout you believe is best for a graniph T-shirt. You can consider brand identity, printing feasibility, and your personal design expertise.”

We then compiled the categories chosen by each designer for later analysis (i.e., comparison with Gemini’s outputs and with actual product layouts). Next, for several representative illustrations in each category, we presented the text explanations generated by Gemini — “Why this layout was chosen” — to multiple illustrators and asked them to rate them based on the 3 criteria (Reasonable / Partially reasonable / Not Reasonable). Designers reviewed each explanation and indicated whether they considered it reasonable; if they considered it unreasonable, they could leave comments. Through this question, we aimed to determine whether the explanation felt plausible or correctly captured the illustrations’ characteristics. Additionally, we recorded the actual placement of each illustration from its commercialized graniph T-shirt (the “production” layout) to use as a “correct” reference for comparison. We

Table 1: Number and percentage of matches between Production layouts and the 5 human designers, based on small categories

Respondent	Matches with production	Match rate with production (%)	Matches with Gemini	Match rate with Gemini (%)
Gemini	12	40.0	-	-
p1	16	53.3	12	40.0
p2	17	56.7	9	30.0
p3	13	43.3	13	43.3
p4	17	56.7	11	36.7
p5	13	43.3	10	33.3

Table 2: Number and percentage of matches between Production layouts and the 5 human designers, based on large categories

Respondent	Matches with Production	Match rate with production (%)	Matches with Gemini	Match rate with Gemini (%)
Gemini	20	66.7	-	-
p1	22	73.3	24	80.0
p2	24	80.0	22	73.3
p3	16	53.5	18	60.0
p4	23	76.7	22	73.3
p5	19	63.3	19	63.3

then analyzed how closely Gemini’s choices (and the choices of the designers in this study) aligned with those real-world, previously adopted layouts.

3.4 Metrics and Analysis Plan

First, we compare the agreement rates of the eight precise categories (i.e., layout definitions that strictly distinguish between front-small, front-middle, etc.) and the five broader categories (grouping similar placements under labels such as “front,” “angled-lower,” etc.). Next, we tabulate the number of matching layout choices among Gemini, the human designers, and the actual product layouts, and evaluate the relationship using Cramér’s V coefficient. Alongside quantitative measures, we also conduct qualitative analyses: specifically, we examine the proportion of illustrations for which designers deemed Gemini’s explanations “reasonable,” identify the characteristics of illustrations that received high ratings, and look at examples that received lower ratings to explore possible reasons.

4 RESULTS

This chapter outlines: (1) the degree of layout agreement with the actual product layouts (production) as a baseline, (2) the statistical validation, and (3) how the explanations provided by Gemini were evaluated in terms of their perceived validity. Detailed figures and comments are available in the Appendix.

4.1 Layout Agreement Analysis

We first examined how often Gemini’s proposed layouts matched the actual T-shirt layouts (“production”) in eight strict categories (front-small, point, etc.). As shown in ¥1¥2 ¥4, Gemini’s match rate was 40% (12/30), whereas the five professional designers ranged from 43% to 57%.

Because “point” and “front-small” are treated separately—despite being quite similar placements—overall agreement tends to decrease. For example, participant p3 often prefers more daring placements such as “front-large” or “back,” resulting in a relatively high match rate with Gemini (13 matches, 43.3%). However, p3 also frequently differed from the other designers.

By contrast, when we grouped these eight categories into five broader ones (front, large front/back, angled-lower, neck-area, other), the agreement rose to 66.7% for Gemini and up to 80% for some designers. ¥1¥2 ¥4 illustrates the improvement in match rates under these broader categories, highlighting that subtle differences like “point” vs. “front-small” were significant factors in strict categorization mismatches.

This result suggests that Gemini’s “visual” layout logic aligns with human-designed layouts at a higher level, even if the AI does not fully account for brand-specific distinctions.

4.2 Statistical Validation

We performed a cross-tabulation between whether Gemini’s choice matched the production layout (“correct/incorrect”) and how many designers (out of 5) also chose that same correct layout (0–5). We then calculated Cramér’s V (¥1¥2 ¥4, Figure 2). The result was a moderate correlation of about $V=0.52$, suggesting that in many cases when Gemini’s selection was correct, multiple designers also selected the correct layout.

4.3 Explanation Validity

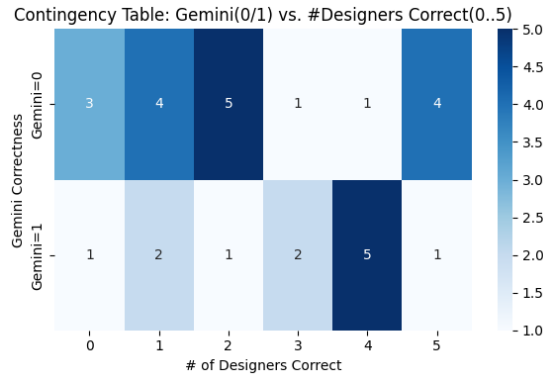
To examine how designers perceived Gemini’s textual explanations, we collected ratings (Yes / Partially Yes / No) across multiple T-shirt illustrations. As shown in Figure 5, approximately 70% of the explanations were considered “reasonable” or “mostly reasonable,”

Table 3: Cross-tabulation of Gemini’s match with production (rows: 0 = mismatch, 1 = match) vs. the number of designers who matched production (columns: 0–5)

Gemini / #Designers Correct	0	1	2	3	4	5	Row Total
Gemini = 0 (Mismatch)	3	4	5	1	1	4	18
Gemini = 1 (Match)	1	2	1	2	5	1	12
Column Total	4	6	6	3	6	5	30

Table 4: Explanation Validity Evaluation

Image ID	#Yes	#Partial	#No	Comment Example
019000752.001.-_1.jpg	3	2	0	“Although the pattern itself might work on the back, if there’s nothing on the front, it effectively becomes a plain T-shirt when coordinating outfits—which might not appeal to customers who prefer printed T-shirts. Also, a large English print on the back is reminiscent of a typical souvenir T-shirt found in tourist areas. Considering the ‘mendako’ (Japanese stubby squid) character theme, it might look like an aquarium souvenir T-shirt. Perhaps placing the design near the hem or disrupting it in some way could achieve a more stylish result.”
019000764.001.-_1.jpg	4	1	0	“If you use a white T-shirt, it may end up looking sparse, so a point layout or large layout might work better. However, depending on the T-shirt color, a small front layout might still be suitable.”
101001418.001.-_1.jpg	3	1	1	“I think it’s an homage to a band T-shirt, so placing a large design on the front or back would work well.”

**Figure 2: Heatmap visualization of the cross-tabulation (Gemini = 0/1 × Number of designers correct = 0–5)**

while the remaining 30% were viewed as “not reasonable” (see Appendix 4.2).

We further analyzed comment logs from the designers. The positive responses often cited that the AI’s reasoning was “visually consistent” or “aligned with their own intuition.” Negative or partially negative evaluations arose when the explanation lacked references to brand identity or printing feasibility. ¥1¥2 ¥4 provides sample comments illustrating designer perspectives. We found that high-rated explanations involved clear references to visual balance

and illustration features, whereas low-rated ones lacked context on brand identity or production feasibility.

Discussion

When we consider the strict classification of eight small categories, the agreement rate between Gemini and the production layouts is around 40%. However, when those categories are grouped into four larger categories, the agreement rate increases to 66.7%. This suggests that, at a broader level, Gemini’s sense of layout is relatively close to human judgment. In comparison with the five designers, Gemini’s small-category alignment is about 10–15 illustrations, and the Cramér’s V of approximately 0.52 indicates at least a moderate correlation.

One reason for the frequent mismatches under the eight-category classification is that brand-specific preferences and practical considerations are not fully integrated into the AI. For example, although “point” and “front-small” might look similar when the shirt is worn, there are subtle differences in actual print size, placement, and a brand’s typical style. Embroidery materials, for instance, are often placed at the “point” position, whereas “front-small” makes up a relatively small proportion of graniph’s product lineup. In other words, the AI primarily focuses on visual elements of the illustration (e.g., shape, color) and may not sufficiently account for brand context or overall product-line consistency. Because Gemini has not been trained on each brand’s past designs, sales data, or technical constraints, a gap arises between “visual optimization” and “real-world business optimization.” When grouped

into larger categories—such as simply calling a smaller print on the front “front” and a larger print “big”—the influence of these small distinctions is reduced, leading to greater overall agreement.

Moreover, the fact that over 70% of Gemini’s layout explanations were judged “reasonable” highlights the value of Explainable AI in creative fields. When designers or illustrators understand why a layout might be optimal, they perceive the AI’s output not just as a “black-box outcome” but as an “idea from a collaborative partner.” In particular, AI’s remarks on aspects like “movement” or “dynamism” often coincide with the designers’ own implicit considerations, fostering a shared vocabulary. On the other hand, these explanations only verbalize visual aspects; there is a lack of details regarding brand-specific concepts or the deeper narrative of the illustration. This likely stems from the AI’s limited awareness of a brand’s past products or printing constraints. By allowing AI to learn these contextual factors in the future, we can expect even more convincing explanations.

Furthermore, these findings highlight the critical importance of merging the generative model’s inherent “rapid idea-generation capabilities” with the human expertise in brand context and knowledge. In particular, for AI researchers, training generative models on brand histories and sales data could extend beyond simple visual optimization and yield proposals that are more closely aligned with real-world operational needs. From the viewpoint of Explainable AI (XAI), the study also reaffirms that clarifying “why this layout was chosen” can facilitate expert-AI collaboration, even in creative fields. To advance AI as a powerful creativity support tool, it is thus essential to integrate not only visual information but also brand concepts and production constraints—treated as meta-information—in a way that enhances explainability.

Overall, these findings suggest that it could be viable for AI to generate numerous rough layout concepts in the early design stages, leaving designers to refine, select, or discard them. Subtle distinctions such as “point” vs. “front-small” can ultimately be adjusted by the designer, while the AI can quickly present a broad range of options. Even if the brand rejects certain ideas—for instance, because they “don’t match our style” or “don’t make sense in terms of cost”—the AI’s explanations could help spark new modifications or variations.

5 LIMITATIONS AND FUTURE DIRECTION

This study has several notable limitations. First, our dataset was restricted to 30 T-shirt illustrations, which may not fully capture the broader variability of designs or brand aesthetics. Second, the AI model (Gemini) did not incorporate brand-specific sales data or printing constraints (e.g., production costs, fabric type, or manufacturing processes). As a result, some proposed layouts may not align well with real-world operational requirements. Third, only five designers participated in the evaluation; while their feedback was valuable, this sample may not reflect the full spectrum of industry professionals, whose preferences could vary by background, cultural context, or personal style. Moreover, regarding Explainable AI (XAI), we relied primarily on questionnaire-based assessments of textual explanations, without an interactive editing environment. This limits our understanding of how designers might actually collaborate with or refine AI-generated proposals in practice.

Looking ahead, several avenues for future research and development emerge. To enhance the model’s applicability, researchers could expand the dataset to include a larger, more diverse range of T-shirt (and potentially other apparel) designs, and train the AI to incorporate brand-specific guidelines and market constraints. It would also be beneficial to produce limited runs of AI-suggested layouts, measure actual sales or user feedback, and feed these real-world results back into the AI to refine its recommendations. Another promising direction involves developing interactive design tools that allow designers to iteratively critique AI outputs and prompt the AI to generate improved alternatives. Such collaborative systems could significantly bolster creativity and efficiency in design workflows. Ultimately, broadening the scope to incorporate brand identity, consumer trends, and dynamic market data may further advance AI’s role in supporting professional fashion design.

6 Conclusion

This study compared AI-generated T-shirt layouts with actual commercial layouts and designs chosen by five professional designers. While agreement was moderate under eight strict layout categories, merging them into five broader categories increased the alignment to around 60–70%. Additionally, many of the AI’s textual explanations were judged “reasonable,” hinting that Explainable AI may facilitate trust in design collaboration.

However, key limitations remain: Gemini did not account for brand-specific or manufacturing constraints, and our sample of five designers may not represent broader industry perspectives. Future work should incorporate larger, more diverse datasets, as well as brand and market data, to enhance real-world applicability. Developing interactive design tools that allow iterative feedback could further strengthen human–AI collaboration. This study lays the groundwork for applying generative AI in fashion design and offers a starting point for more holistic, context-aware systems.

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A APPENDICES

A.1. Example Prompt for Gemini

Below is the prompt used in our experiment:

```
``You are a professional apparel designer for a brand called graniph. Your task is to advise on the optimal placement and size for printing an image on a T-shirt. Taking the provided image into account, choose only one of the following options, considering fairly which you believe to be the most beautiful and effective arrangement and size. Also, please explain why you selected that option.
```

```
~~~~
```

Placement and Size Options:

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```

Chest (small, left side) -- Ideal for simple logos, icons, or pocket-area designs.

Front-small (width/height around 10cm, center of the front) -- Best for smaller designs or single-point accents.

Front-medium (width/height around 15--20cm, center of the front) -- Suited for medium-sized designs or text.

Front-large (height of 25cm or more, center of the front) -- Ideal for bold designs or those aiming to make a strong impact.

Angled-lower (positioned diagonally near the lower part of the T-shirt) -- Suited for designs that convey movement or a unique placement.

Shoulder/neck area (around the shoulders or near the neckline) -- Great for a subtle design or accent detail.

Back (on the back of the T-shirt) -- Suitable for placing a message or design on the back.

Other -- Any arrangement not covered above.

Please refer to this image for guidance: {Attach an image illustrating layout options}

```
~~~~
```

Please provide your answer in the following format:

Selected option: [Number of the option you chose]

[Please give a detailed explanation of why this option is optimal, taking into account the nature of the image.]''

Additionally, for the Gemini API call, we specified the following parameters:

```
generation\_config = {
    "max\_output\_tokens": 8192,
    "temperature": 1.2,
    "top\_p": 0.8,
    "response\_mime\_type": "application/json",
    "response\_schema": {"type": "OBJECT", "properties": {"option": {"type": "INTEGER", "description": "option id"}, "reason": {"type": "STRING", "description": "Why this option is sufficient"}}, "required": ["option", "reason"]},
}
safety\_settings = [
    SafetySetting(
        category=SafetySetting.HarmCategory.HARM\_CATEGORY\_HATE\_SPEECH,
        threshold=SafetySetting.HarmBlockThreshold.OFF
    ),
    SafetySetting(
        category=SafetySetting.HarmCategory.HARM\_CATEGORY\_DANGEROUS\_CONTENT,
        threshold=SafetySetting.HarmBlockThreshold.OFF
    ),
    SafetySetting(
        category=SafetySetting.HarmCategory.HARM\_CATEGORY\_SEXUALLY\_EXPLICIT,
        threshold=SafetySetting.HarmBlockThreshold.OFF
    ),
    SafetySetting(
        category=SafetySetting.HarmCategory.HARM\_CATEGORY\_HARASSMENT,
        threshold=SafetySetting.HarmBlockThreshold.OFF
    ),
]
```

A.2 Additional Figures and Tables

This section provides figures and tables that could not be included in the main text due to length and layout constraints

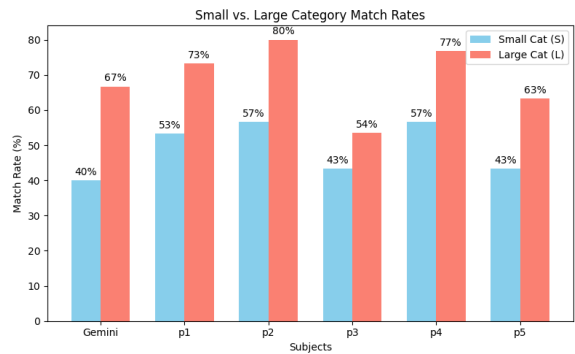


Figure 3: Comparison of Gemini vs. product match rates in small vs. large categories

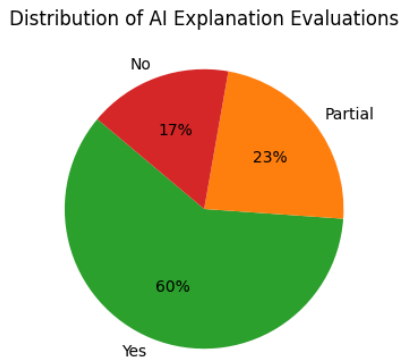


Figure 4: Distribution of Explanation Validity Ratings (Yes / Partially / No)

A.3 Samples of Generated Explanations

Below are example explanations from Gemini, describing why a particular layout was chosen.

- 101000916.100.-_1 (Option 7: Back) : “This design contains a lot of visual information and is vertically oriented. Placing it large on the back makes it easier to see the details of the artwork, and it allows viewers to fully appreciate the distinctive mood created by the combination of the illustration and the German text. By leaving the front blank, the back design stands out more, giving the impression that the wearer is carrying the ‘secret of the sea’ on their back, creating a sense of mystery. In addition, compared to other choices, using the back offers a higher degree of freedom, making it possible to best utilize the unique atmosphere of this image.”
- 101001302.001.-_1 (Option 2: Front-Small) : “This penguin illustration is simple and cute. If we placed it small on the chest (option 1), the charm of the illustration would be underemphasized. On the other hand, if we went with front-medium or larger, it would look empty because the design is so simple, potentially unbalancing the T-shirt overall. By choosing front-small, we preserve the penguin’s charm while giving the T-shirt a moderate presence. Center placement naturally guides the eye and achieves a balanced finish. It also aligns well with graniph’s brand image of understated stylishness. Compared with the other options, front-small best brings out the characteristics of this illustration.”
- 101001418.001.-_1 (Option 1: Chest Point) : “I considered the graniph brand image, which emphasizes subtle, stylish design. While a tiger illustration naturally has a strong presence, placing it large on the front might make it overwhelming. By positioning it small on the chest, you can create an elegant and sophisticated look that appeals to a wide range of ages. Placing it on the left chest is an effective accent that does not interfere with coordinating other items. Keeping the size small prevents the illustration’s details from being lost, allowing the delicate linework to remain visible.”

4. 019000916.417.-_1 (Option 5: Angled-Lower) : “To make the red panda appear as if it’s clinging to the T-shirt, thereby highlighting the humorous and dynamic aspect of the design, I chose the angled-lower placement. Positioning it on the chest or front would not fully convey the appeal of the red panda’s pose. By placing it angled-lower, it looks as though the red panda is peeking from the hem of the T-shirt, emphasizing movement and catching the viewer’s eye. The size is neither too small nor too large, allowing the red panda to stand out while keeping the overall balance and projecting a sense of cuteness. Moreover, this layout matches the playful design aesthetic characteristic of graniph and helps set it apart from other brands. Rather than defaulting to a central placement like options 3 or 4, which might feel conventional, I determined that angled-lower best leverages the dynamic quality of the illustration.”
5. 101000916.100.-_1 (Option 3: Front-Medium) : “This image features four cat faces drawn large for strong visual impact. Therefore, placing it at a moderately large size on the front is effective. With option 3 (front-medium), the cats’ expressions and details remain visible, while the size is not overly large, maintaining a good balance. It has enough presence to be eye-catching while still being appropriate for everyday wear. Considering the design’s impact, the expression of details, graniph’s brand image, and the versatility of possible wearing situations, I decided that option 3 (front-medium) is the most suitable choice.”